Topic Test 5 · Paper 1

# Question 1

How many positive real roots does

$$f(x) = x^4 - 8x^3 + 22x^2 - 24x$$

have?

**A** 0

**B** 1

**C** 2

**D** 3

**E** 4

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# Question 2

In the expansion of $(2x + b)^9$, the coefficient of $x^5$ is equal to the coefficient of $x^3$ in $(3b + x)^6$.

Given that $b > 0$, find $b$.

A $\dfrac{15}{112}$

B $\dfrac{112}{15}$

C $\dfrac{15}{56}$

D $\dfrac{56}{15}$

E $\dfrac{5}{42}$

Topic Test 5 · Paper 1

# Question 3

Consider the tangent to the curve $y = x^2 + bx$ at $x = 2$.

For which values of $b$ is the $x$-intercept of the tangent greater than $4$?

A $-3 < b < 3$

B $3 < b < 4$

C $b > 3$

D $-4 < b < -3$

E $b > -3$

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# Question 4

The line $y = mx + 1$ is tangent to the circle

$$(x - 2)^2 + (y + 1)^2 = 5.$$

There are two possible values of $m$. What is their sum?

**A** $-8$

**B** $-1$

**C** $1$

**D** $8$

**E** $16$

Topic Test 5 · Paper 1

# Question 5

Suppose

$$5^{2+5+8+\cdots+(3x-1)} = 0.04^{-50},$$

where $x$ is a positive integer. Find $x$.

**A** 5

**B** 6

**C** 7

**D** 8

**E** 9

Topic Test 5 · Paper 1

# Question 6

Three distinct cards are chosen uniformly at random from cards numbered $1, 2, \ldots, 7$.

Given that the product of the three chosen numbers is even, what is the probability that their sum is even?

A $\frac{12}{31}$

B $\frac{15}{31}$

C $\frac{18}{31}$

D $\frac{19}{31}$

E $\frac{20}{31}$

Topic Test 5 · Paper 1

# Question 7

Which of the following is a line of symmetry of the graph

$$y = \frac{1}{\sin\left(4x + \frac{\pi}{3}\right)}?$$

A $x = \dfrac{13\pi}{2}$

B $x = \dfrac{\pi}{2}$

C $x = \pi$

D $x = \dfrac{13\pi}{24}$

E $y = \dfrac{5\pi}{24}$

Topic Test 5 · Paper 1

# Question 8

A sequence is defined by

$$x_{n+1} = \begin{cases} \dfrac{x_n}{2}, & x_n \text{ even}, \\ \dfrac{3x_n + 1}{2}, & x_n \text{ odd}. \end{cases}$$

Given $x_1 = 37$, find $x_{2026}$.

**A** 1

**B** 2

**C** 4

**D** 5

**E** 8

Topic Test 5 · Paper 1

# Question 9

How many solutions does

$$\cos(2x)\ln x = \sin(2x)$$

have in the interval $0 < x < 3\pi$?

**A** 0

**B** 3

**C** 5

**D** 6

**E** 7

Topic Test 5 · Paper 1

# Question 10

Let $f$ be integrable on $[-4, 4]$ and satisfy

$$f(x) + f(-x) = x^2$$

for all $x \in [-4, 4]$. Given that

$$\int_0^4 f(x)\,dx = 3,$$

evaluate

$$\int_{-2}^{2} \big[f(x-2) - f(2-x) + 1\big]\,dx.$$

A $\dfrac{46}{3}$

B $\dfrac{49}{3}$

C $\dfrac{55}{3}$

D $\dfrac{58}{3}$

E $\dfrac{64}{3}$

Topic Test 5 · Paper 1

# Question 11

The curves

$$y = \pi x(x-1)(x-2)(x-3)(x-4)(x-5)$$

and

$$y = \frac{5}{17}x + \frac{\pi}{3}$$

intersect at six points. Find the sum of the six $x$-coordinates.

**A** $-15$

**B** $-\dfrac{44\pi}{51} + 1$

**C** $-\dfrac{7}{15}$

**D** 0

**E** $\dfrac{7}{15}$

**F** $\dfrac{44\pi}{51} - 1$

**G** 12

**H** 15

Topic Test 5 · Paper 1

# Question 12

A monic cubic function $f$ satisfies

$$f(1) = 1, \qquad f(2) = 1, \qquad f(-1) = -5.$$

Find $f(3)$.

**A** $-7$

**B** 0

**C** 4

**D** 6

**E** 7

**F** 36

Topic Test 5 · Paper 1

# Question 13

Integers $a, b, c$, each greater than $1$, satisfy

$$abc + ab + bc + ca + a + b + c = 104.$$

Find $a + b + c$.

**A** 10

**B** 11

**C** 12

**D** 13

**E** 14

**F** 15

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# Question 14

Let

$$u_n = 3u_{n-1} + 2u_{n-2}, \qquad u_1 = 1, \quad u_2 = 5.$$

Find

$$\lim_{k \to \infty} \frac{u_k}{u_{k-1}}.$$

**A** 2

**B** $\sqrt{5}$

**C** $\dfrac{3 + \sqrt{17}}{2}$

**D** $\dfrac{3 \pm \sqrt{17}}{2}$

**E** $\dfrac{3 + \sqrt{5}}{2}$

**F** $1 \pm \sqrt{2}$

**G** $3 + \sqrt{2}$

Topic Test 5 · Paper 1

# Question 15

For every integer $n \geq 0$,

$$\int_{n}^{n+1} f(x)\,dx = n^2 + n + 1.$$

Evaluate

$$\int_{0}^{3} f(x)\,dx + \int_{2}^{5} f(x)\,dx - 2\int_{1}^{4} f(x)\,dx.$$

**A** 2

**B** 4

**C** 5

**D** 6

**E** 8

**F** 10

Topic Test 5 · Paper 1

# Question 16

For each odd integer $n$ from 3 to 25 inclusive, the coefficient of $x^2$ in $(1+x)^n$ is recorded. Add together all values of $n$ whose recorded coefficient is divisible by 12. What is the result?

**A** 18

**B** 24

**C** 26

**D** 34

**E** 42

**F** 48

**G** 51

Topic Test 5 · Paper 1

# Question 17

The graph of $y = \sqrt{5x - 2}$ undergoes the following transformations in order:

1. translation $4$ units left;
2. translation $6$ units down;
3. reflection in the $x$-axis;
4. stretch parallel to the $y$-axis with scale factor $\frac{1}{3}$;
5. stretch parallel to the $x$-axis with scale factor $2$.

Which equation describes the final graph?

**A** $y = \sqrt{2 - \dfrac{5x}{18}} - 2$

**B** $y = \dfrac{\sqrt{-10x - 6} - 4}{3}$

**C** $y = -\dfrac{\sqrt{\frac{5}{2}x - 22} - 6}{3}$

**D** $y = -\dfrac{\sqrt{\frac{5}{2}x + 18} - 6}{3}$

**E** $y = \dfrac{\sqrt{\frac{5}{2}x - 22} - 6}{3}$

**F** $y = -\dfrac{\sqrt{\frac{5}{2}x + 22} - 6}{3}$

**G** $y = \dfrac{\sqrt{\frac{5}{2}x + 14}}{3}$

Topic Test 5 · Paper 1

# Question 18

How many values of $x$ in

$$-2\pi \leq x \leq 2\pi$$

satisfy

$$\sin 3x = \cos 2x?$$

**A** 6

**B** 8

**C** 9

**D** 10

**E** 12

**F** 14

Topic Test 5 · Paper 1

# Question 19

For

$$y = 2\cos 3x + 2\sqrt{3}\sin 3x + 7, \qquad 0 \le x < \pi,$$

find the sum of the $x$-coordinates of all stationary points.

**A** $\pi$

**B** $\dfrac{7\pi}{6}$

**C** $\dfrac{4\pi}{3}$

**D** $\dfrac{3\pi}{2}$

**E** $\dfrac{5\pi}{3}$

**F** $2\pi$

Topic Test 5 · Paper 1

# Question 20

Let $k > 0$, $f_0(x) = x$, and

$$f_{n+1}(x) = |f_n(x) - k|$$

for non-negative integers $n$. For $n > 0$, let $\alpha$ and $\beta$ be the least and greatest real numbers for which $f_n(x) = 0$.

Find

$$\int_{\alpha}^{\beta} f_n(x)\,dx$$

in terms of $n$ and $k$.

A $nk^2$

B $nk^2 - k^2$

C $kn^2$

D $nk$

E $k(n-1)^2$

F $nk^2 - 1$